Synthesis hackathon: every partner bounty verified

The Curated LP project can credibly target 8 bounties worth $34,250 total — but three critical gaps will break your submission if unaddressed. The Uniswap Trading API does not yet support LP operations (marked "Coming Soon"), ERC-8004 contracts are not deployed on Base mainnet, and Bonfires.ai has no public API. Every other integration path has been verified against live documentation and on-chain data as of March 13, 2026. Below is the complete partner-by-partner technical specification with zero assumptions.

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1. Uniswap ($5,000): the API key is mandatory, hooks are bonus

The bounty title — "Best Uniswap API Integration" — means exactly what it says. The **Uniswap Trading API** is a REST service at `https://trade-api.gateway.uniswap.org/v1/` that requires an API key generated at `developers.uniswap.org/dashboard`. Authentication uses an `x-api-key` header plus `x-universal-router-version: 2.0`. Uniswap The API routes swaps through v2, v3, v4 pools and UniswapX. Uniswap

The critical gap: LP operations via the Trading API are listed as "Coming Soon." The exact quote from api-docs.uniswap.org: "*(Coming Soon) Liquidity Providing (LPing) — Create, modify, and remove liquidity positions from Uniswap protocol pools*." Uniswap This means your AI agent cannot manage concentrated liquidity positions through the API. For LP management, you must interact directly with the **PositionManager** contract at `0x7c5f5a4bbd8fd63184577525326123b519429bdc` on Base. uniswap

The scoring structure is tiered. The minimum requirement is a working API integration with real swap TxIDs on testnet or mainnet, an open-source GitHub repo with README, and a real API key. Bonus depth comes from integrating additional Uniswap stack layers: v4 hooks, AI Skills, Unichain, v4 contracts, and Permit2.

Uniswap AI Skills (from github.com/Uniswap/uniswap-ai, released February 21, 2026) provide seven agent-compatible tools: `v4-security-foundations`, `configurator`, `deployer`, `viem-integration`, `swap-integration`, `liquidity-planner`, and `swap-planner`. These are TypeScript packages installable via `npx skills add uniswap/uniswap-ai` GitHub or `npm install @uniswap-ai/core`. They ship as Claude Code plugins but are explicitly agent-agnostic. github The repo also includes a `uniswap-hooks` plugin with an `aggregator-hook-creator` skill for building v4 hooks. github

Recommended architecture: The AI curator agent uses the Trading API (with real API key) for swap routing and price discovery, uses `liquidity-planner` and `swap-planner` AI Skills for strategy generation, calls PositionManager directly for LP operations via the v4 hook, and the custom hook implements `beforeAddLiquidity`/`afterSwap` callbacks on the Base PoolManager.

Verified Base v4 deployment addresses:

Contract	Address
PoolManager	0x498581ff718922c3f8e6a244956af099b2652b2b
PositionManager	0x7c5f5a4bbd8fd63184577525326123b519429bdc
Universal Router	0x6ff5693b99212da76ad316178a184ab56d299b43
Quoter	0x0d5e0f971ed27fbff6c2837bf31316121532048d
StateView	0xa3c0c9b65bad0b08107aa264b0f3db444b867a71
Permit2	0x000000000022D473030F116dDEE9F6B43aC78BA3

Unichain (chain ID 130) is Uniswap Labs' own OP Stack L2 — separate from Base. uniswap Deploying on Unichain in addition to Base would earn extra bonus points but is not required.

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2. Venice AI ($11,500): OpenAI-compatible with real privacy constraints

Venice's API is **fully OpenAI-compatible**. The base URL is https://api.venice.ai/api/v1, Venice API Docs authentication is Authorization: Bearer VENICE_API_KEY, venice and you can use the standard OpenAI Python or JS SDK with only the base URL and API key changed. Apidog Keys are managed at venice.ai/settings/api with prefixes VENICE-ADMIN-KEY- or VENICE-INFERENCE-KEY-. Venice

The chat completions endpoint (POST /api/v1/chat/completions) supports streaming via SSE. **Function calling is supported** on select models: zai-org-glm-4.7 (128k context, flagship reasoning), qwen3-4b, mistral-31-24b, and llama-3.3-70b. Venice API Docs For financial market analysis, the best choices are zai-org-glm-4.7 for complex reasoning with tool use, Venice API Docs or qwen3-235b for raw analytical power. deepseek-ai-DeepSeek-R1 offers extended chain-of-thought reasoning for multi-step problems.

The venice_parameters object provides eight options beyond the OpenAI spec:

- enable_web_search: "off" | "on" | "auto" — enables real-time web data retrieval
- enable_web_scraping: boolean — scrapes up to 5 URLs from the user message
- enable_web_citations: boolean — LLM cites sources in [REF]0[/REF] format
- strip_thinking_response: boolean — removes <think></think> blocks
- disable_thinking: boolean — disables reasoning on thinking models
- include_search_results_in_stream: boolean — experimental, injects search results as first SSE chunk
- return_search_results_as_documents: boolean — LangChain-compatible document returns
- include_venice_system_prompt: boolean — toggles Venice's default system prompts venice Venice

"Private cognition" is not encrypted inference. Venice explicitly states homomorphic encryption is aspirational and not yet feasible. Venice Their privacy architecture means: zero server-side storage or logging of prompts/responses, Skywork SSL encryption in transit, transient GPU processing with immediate purge, Venice and no identifying information sent to GPU infrastructure. Skywork During inference, prompts exist momentarily as plaintext on the GPU. Venice This is privacy through data minimization, not cryptographic privacy.

Pricing and access: Pro account gives **$10 API credit** on signup. Venice Venice VVV staking provides proportional inference capacity (1% of staked VVV = 1% of API throughput). Venice No Synthesis-specific free tier was found. API pricing: Venice Small at $0.05/$0.15 per million tokens (input/output), Venice Large at $0.90/$4.50, Venice web search at $10 per 1K calls. Venice

No proprietary SDK exists — use OpenAI SDKs. Community options include venice-ai on PyPI (sync/async with streaming) GitHub and venice-cli (official Node.js CLI). GitHub GitHub Framework integrations exist for LangChain, Mastra, and Eliza. Mastra

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3. MetaMask ($5,000): delegation framework is live on Base and perfect for this use case

The MetaMask Delegation Framework is an **on-chain permission system** that separates account ownership from execution. A delegator (LP depositor) signs an off-chain delegation granting specific powers to a delegate (AI curator agent). The delegation is cryptographically signed, stored anywhere (IPFS, cloud, local), and only goes on-chain when the delegate redeems it through the DelegationManager contract.

This is precisely what the Curated LP vault needs. The architecture maps directly: depositor's DeleGator smart account (ERC-4337 compliant) is the delegator, the AI agent's account is the delegate, and caveats constrain what the agent can do — call specific DEX functions, stay within spending limits, operate within time windows. The agent redeems the delegation via the DelegationManager to rebalance positions on the depositor's behalf.

The delegation data structure includes: delegate (recipient address), delegator (granting address), authority (parent delegation hash or ROOT_AUTHORITY), caveats (restriction array), salt, and signature. Metamask Sub-delegations create chains — Bob can redelegate Alice's authority to Carol with tighter restrictions. Revoking the root delegation invalidates all downstream delegations instantly.

ERC-7710 defines how delegations are redeemed on-chain (the redeemDelegations function interface). **ERC-7715** defines how dapps request permissions from wallets (wallet_grantPermissions JSON-RPC method). MetaMask Together: ERC-7715 handles the wallet-to-dapp communication flow, ERC-7710 handles on-chain enforcement.

Confirmed live on Base mainnet with Delegation Framework v1.3.0 contracts. Also live on Ethereum, Polygon, BSC, Optimism, Arbitrum, Linea, and Gnosis Chain. Base Sepolia supported for testing. Contract addresses are resolvable via getSmartAccountsEnvironment(base.id) in the SDK. Note: ERC-7715 Snaps features work only on Sepolia currently, but core delegation functionality works on all supported mainnets.

The toolkit is now called **@metamask/smart-accounts-kit** (v0.3.0), renamed from @metamask/delegation-toolkit. It's a Viem-based TypeScript library. Scaffold via npx @metamask/create-gator-app with templates

including "MetaMask Smart Accounts & Delegation Starter." Metamask

"**Intent-based delegations**" means the delegation expresses user intent (e.g., "optimize my LP position") while caveats constrain execution boundaries, giving the agent freedom in *how* it fulfills the intent. "**ZK proofs with delegation-based authorization**" would require custom caveat enforcer contracts that verify zero-knowledge proofs before allowing delegation redemption — a novel research direction, not a built-in feature. "**Extend ERC-7715 with sub-delegations**" means building hierarchical permission trees with custom caveat enforcers, which the framework explicitly supports: "*Developers can build new Caveat Enforcers for their own use cases, and the possibilities are endless*." GitHub

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4. Merit Systems ($5,250): x402 is HTTP-native payments, AgentCash is the distribution layer

x402 is an open-source protocol by Coinbase that revives HTTP status code 402 ("Payment Required") for machine-to-machine stablecoin payments. Circle The flow: client sends a normal HTTP request → server responds with `402 + PAYMENT-REQUIRED header` → client signs a USDC payment → client retries with `X-PAYMENT header` → server verifies payment → server returns `200 OK` with the resource. LabLab Payment uses **EIP-3009 (transferWithAuthorization)** for gasless USDC transfers. LabLab Circle

x402 is live on Base (eip155:8453) with USDC as the primary token. DEV Community Medium Also supports Polygon and Solana. The CDP facilitator at `x402.coinbase.com` offers **1,000 free transactions per month**, then $0.001/transaction.

AgentCash is Merit Systems' product (launched March 10, 2026) that gives agents a single USDC balance to access **280+ x402-protected API endpoints**. X It ships as **Tessl Skills** (SKILL.md files), not a traditional MCP server. Install via `npx tessl i github:Merit-Systems/agentcash-skills --skill email`. Tessl The execution primitive is `agentcash.fetch(url, method, body)` which auto-handles the 402→pay→retry flow. Routes span social intelligence (Twitter/X via Grok, Reddit), web research (Exa neural search, Firecrawl), email, file hosting, and image/video generation across Merit's `*.stableenrich.dev` and `*.stableemail.dev` domains. LobeHub

Consuming x402 APIs (client-side): Three approaches — `@x402/axios` wrapper (auto-handles 402 flow), `agentcash.fetch` CLI, or `x402-fetch` npm package. For MCP servers, the x402 client detects 402 responses, handles payment automatically, and returns paid data. Coinbase

Producing x402 endpoints (server-side): One middleware line:

```javascript
import { paymentMiddleware } from "x402-express";
app.use(paymentMiddleware({
  price: "$0.001",
  network: "eip155:8453",
  facilitatorUrl: "https://x402.coinbase.com"
}));
```

For the Curated LP project, "load-bearing, not decorative" means the x402 payment layer must be integral to the agent's function. Example: the curator agent pays for real-time market data via x402 to make LP decisions,

and/or the vault charges performance fees as x402-gated API access to strategy analytics. The GitHub repo is at github.com/coinbase/x402 (Apache-2.0, 5.4k stars). (github) (GitHub)

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5. Valory/Olas ($3,000): three distinct tracks, all work on Base

Olas runs three standard hackathon bounty tracks, and **all three work on Base** — the mech-client documentation explicitly lists Base as a supported chain with Marketplace contracts, Agent Mode, Native Payment, OLAS Payments, and USDC Payments. (olas)

"Hire an Agent" ($1,000) — Demand Side: Use mech-client (`pip install mech-client`, requires Python ≥3.10, <3.12) to consume Mech agent services. (olas) Your application must make **at least 10 requests** to Mechs via the Marketplace, demonstrating meaningful usage. Usage: `mechx request --prompts "your prompt" --priority-mech <address> --tools <tool> --chain-config base`, or via Python: `from mech_client.marketplace_interact import marketplace_interact`. (olas) (Olas) Payment supports native token, OLAS, and USDC. (olas)

"Monetize Your Agent" ($1,000) — Supply Side: Build and deploy a custom Mech using the MechKit repository (github.com/valory-xyz/mech). Your deployed Mech must serve **at least 50 requests** from other agents/applications. The technical flow: requester writes to IPFS → Mech listens for on-chain requests → performs off-chain tasks → delivers results on-chain. (Olas)

"Build an Agent for Pearl" ($1,000): Pearl is the "AI Agent App Store" — a desktop app (Windows/macOS/Linux) for running autonomous agents. (CoinDesk) (Pearl) **Pearl integration is currently access-restricted to Accelerator participants** per the stack.olas.network documentation. This track requires a standalone executable binary, Docker image, and following specific SDK specs. (Olas) The QA checklist is referenced in the Agent Integration Checklist at olas.network/accelerator.

Can you target multiple sub-bounties? The tracks are evaluated independently. Building a project that qualifies for both demand-side (consuming Mechs) and supply-side (providing a Mech) is theoretically possible but ambitious. The Pearl track's access restriction makes it the riskiest of the three.

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6. Locus ($3,000): MCP server confirmed, USDC-on-Base only

Locus (YC F25) is the payment layer for autonomous AI agents. (Locus) Founded by ex-Coinbase and ex-Scale AI engineers, it provides non-custodial smart wallets on Base with programmatic spending controls. (Y Combinator)

Setup flow: Create account at app.paywithlocus.com → deploy wallet → create an agent identity → set permissions via policy groups → generate API key. (n8n)

MCP server confirmed at `https://mcp.paywithlocus.com/mcp` with Bearer authentication using your Locus API key. (n8n) Compatible with n8n, Claude, and other MCP clients. A published Locus skill for Claude exists at clawd.bot. (LinkedIn)

Spending controls operate through a policy engine with: per-transaction limits (e.g., max $50 per payout), daily spending caps (e.g., $500/day), required transaction justifications, (Y Combinator) agent identities bound to policy

groups, approval flows, and full audit trails. LinkedIn Y Combinator

"USDC on Base only" is confirmed multiple times in official sources. Their YC launch page states: "*Locus currently supports USDC on Base, with support for ACH and wire transfers coming soon.*" Y Combinator Y Combinator They also built a reference implementation of OpenAI's Agentic Commerce Protocol (ACP) at github.com/locus-technologies/agentic-commerce-protocol-demo. GitHub

"Automatic disqualification without working integration": While the exact bounty text was not captured from the dynamically-loaded Devfolio page, a "working integration" would mean an agent that successfully deploys a Locus wallet, executes USDC transactions on Base, and demonstrates spending controls via the API/MCP.

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7. Self Protocol ($1,000): five registration modes with MCP server

Self Agent ID registers AI agents with passport-verified identity. Self From app.ai.self.xyz: "*Register your AI agent with a passport scan. Choose from five modes — wallet-based, agent keypair, wallet-free, passkey smart wallet, or social login via Privy. Your agent gets a soulbound NFT and an A2A-compatible identity card that any service can verify instantly.*" Self

SDKs in three languages: `@selfxyz/agent-sdk` (npm), `selfxyz-agent-sdk` (pip), `self-agent-sdk` (cargo). **MCP server**: `@selfxyz/mcp-server` — quote: "*Building with AI agents? Use the MCP server to give your agent identity directly from your IDE.*" Self

The **soulbound NFT** is a non-transferable ERC-721 token minted during registration, serving as the agent's on-chain identity credential. It's built on ERC-8004's Identity Registry pattern (an upgradeable ERC-721 with `register` minting and `agentURI` metadata). PayRam Buildbear **A2A verification** follows Google's Agent-to-Agent protocol, where agents discover each other via Agent Cards (JSON metadata documents advertising identity, capabilities, and authentication requirements). Ping Identity

Self Protocol supports **18+ EVM chains** including Base, Ethereum, Polygon, Arbitrum, and Optimism. Chainstack Celo has the deepest integration (co-sponsored ZK audit, ODIS infrastructure), Zksecurity but Base is supported. Each identity document maps to a unique identifier — one person cannot register unlimited agents. Self

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8. ENS ($1,500): Basenames are fully ENS-compatible on Base

Three sub-tracks: Identity ($600), Communication ($600), Open Integration ($300). **Basenames (*.base.eth)** are fully onchain, leveraging ENS infrastructure deployed on Base. Base The architecture has three components: an L1 ENS resolver on Ethereum for the `base.eth` second-level domain, an ENS-like registry/registrar/resolver on Base for `*.base.eth` subdomains, and an off-chain CCIP gateway (ENSIP-10) bridging the two. GitHub

Resolution works both ways on Base. Forward resolution (name → address) works cross-chain via the CCIP gateway. ENS Reverse resolution (address → name) is supported via **ENSIP-19 L2 Primary Names**, working

on Arbitrum, Base, Linea, OP Mainnet, and Scroll. ENS Any ENS-compliant client resolves Basenames without Base-specific code. Paragraph

On-chain resolution from a hook: A v4 hook could resolve Basenames by reading from Base's L2 resolver contracts directly. The L2ResolverAbi with name function at BASENAME_L2_RESOLVER_ADDRESS is callable from Solidity. Base This enables ENS-based identity for curator agents (Identity track), agent-to-agent communication via ENS text records (Communication track), and novel hook-level ENS integration (Open Integration track).

Can one project target all three? Hackathon judges typically award a project to the single best-fitting sub-track. The three tracks are complementary, so a strong integration could potentially be considered across all three, but conservatively plan for one $600 award.

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9. Bonfires.ai ($3,000) and Arkhai ($900): highest-risk integrations

Bonfires.ai is a knowledge graph platform for AI agents — "*Explore, visualize, and interact with knowledge graphs. Create data rooms, generate hyperblogs, and leverage AI agents for intelligent discovery.*" Bonfires It uses Irys for data storage X and has a $KNOW token. X **No public API or SDK was found.** The documentation at publish.obsidian.md/bonfires requires JavaScript rendering and returned blank. Without a documented API, integration during a hackathon would require either the Bonfires.ai platform UI or direct coordination with their team. **This is a $3,000 bounty with significant integration risk due to the API gap.**

Arkhai/Alkahest provides programmable marketplace primitives: escrow smart contracts with Boolean logic, recursive arbitration, and natural language agreements. SDKs exist in Python (alkahest-py), Rust (alkahest-rs), and TypeScript. **However, the Python SDK was archived on February 14, 2026** — one month before the hackathon. This could indicate the SDK has been superseded or consolidated. The Python SDK supports AlkahestClient with submodules for ERC20/ERC721 operations, escrow creation, and arbiter configuration. GitHub A performance fee mechanism using Alkahest escrow is architecturally viable but the archived SDK raises reliability concerns. **At $900, the effort-to-reward ratio is questionable given the SDK uncertainty.**

Contract verification: four confirmed, two failed

Contract	Address	Status
Uniswap v4 PoolManager (Base)	0x498581ff718922c3f8e6a244956af099b2652b2b	✅ Live, official
Uniswap v4 PositionManager (Base)	0x7c5f5a4bbd8fd63184577525326123b519429bdc	✅ Live, official
wstETH (Base)	0xc1CBa3fCea344f92D9239c08C0568f6F2F0ee452	✅ Live, official Lido bridge, ~39,775 wstETH, ~457K holders Base

Contract	Address	Status
Chainlink wstETH/stETH Rate (Base)	0xB88BAc61a4Ca37C43a3725912B1f472c9A5bc061	✅ Live, official — note: this is an exchange rate feed, not a USD price feed Lido
Talent Protocol Builder Score (Base)	0xBBFeDA7c4d8d9Df752542b03CdD715F790B32D0B	⚠️ Live but snapshot-only — only Builder+ members (0.001 ETH) have on-chain scores
ERC-8004 IdentityRegistry (Base)	Not deployed	❌ Official repo lists Base Sepolia as "to be deployed"; Base mainnet not listed GitHub
ERC-8004 ReputationRegistry (Base)	Not deployed	❌ ERC-8004 is still DRAFT status, not finalized

The Talent Protocol contract is callable but limited. getScore() returns results only for users who explicitly paid to snapshot their Builder Score on-chain. Most users won't have an on-chain score. The Talent API is the more complete data source for Builder Score queries.

ERC-8004 is a blocker for any on-chain agent identity registry on Base. The standard is in Draft status, co-authored by engineers from MetaMask, Ethereum Foundation, Google, and Coinbase. GitHub Backpack Exchange Only Ethereum Sepolia testnet has confirmed deployments. GitHub Any plan relying on ERC-8004 registries on Base mainnet must be redesigned or use Self Protocol's agent-sdk instead.

Bounty targeting strategy: recommended 8-bounty configuration

Based on verified integration feasibility, target these bounties in priority order:

1. **Venice AI ($11,500)** — Core: AI curator uses Venice for market analysis with function calling and web search. Highest prize, lowest integration friction (OpenAI-compatible API).
2. **Uniswap ($5,000)** — Core: v4 hook on Base + Trading API for swap routing + AI Skills for strategy generation. Satisfies both API key requirement and hook depth bonus.
3. **MetaMask ($5,000)** — Core: LP depositors delegate concentrated liquidity management to AI agent via DeleGator accounts with custom caveats. Perfect architectural fit.
4. **Merit/x402 ($5,250)** — The curator agent pays for real-time market data via x402 endpoints; vault exposes strategy analytics as x402-gated API. Load-bearing, not decorative.
5. **Locus ($3,000)** — Agent's operational wallet for gas and data costs. Spending controls enforce per-rebalance limits and daily caps. Y Combinator MCP integration for agent wallet management. n8n

6. **Self Protocol ($1,000)** — Register each curator agent with a soulbound NFT identity. Quick win via @selfxyz/agent-sdk or MCP server.
7. **ENS ($1,500)** — Curator agents use Basenames (e.g., vault-curator.base.eth). Depositors resolve agent identity via ENS. Target the $600 Identity sub-track.
8. **Olas ($1,000)** — Use mech-client on Base to consume Mech services for market data or predictions. olas Target the "Hire an Agent" demand-side track with ≥10 requests.

Drop from consideration: Bonfires.ai (no API), Arkhai (archived SDK, low reward), Pearl integration (access-restricted), and the Olas supply-side track (50 requests is a high bar for a hackathon).

Total targetable value: $33,250 across 8 bounties. The Synthesis hackathon uses Devfolio for submissions. Synthesis Based on comparable Devfolio-powered events, projects can typically apply for multiple sponsor prizes. Devfolio No explicit rule against multi-bounty targeting was found in the Synthesis documentation, but the specific Devfolio submission form may limit selections. Join the hackathon chat at nsb.dev/synthesis-chat synthesis and check github.com/sodofi/synthesis-hackathon for the latest rules GitHub — the bounty details were still being rolled out as of today's build start date.